

# CAIP: Detecting Network Device Misconfigurations with Context-Aware Iterative Prompting of LLMs

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## Abstract

Model checkers and consistency checkers detect critical errors in router configurations, but these tools require significant manual effort to develop and maintain. LLM-based Q&A models have emerged as a promising alternative, allowing users to query partitions of configurations through prompts and receive answers based on learned patterns, thanks to transformer models pre-trained on vast datasets that provide generic configuration context for interpreting router configurations. Yet, current methods of partition-based prompting often do not provide enough network-specific context from the actual configurations to enable accurate inference. We introduce a Context-Aware Iterative Prompting (CAIP) framework that automates network-specific context extraction and optimizes LLM prompts for more precise router misconfiguration detection. CAIP addresses three challenges: (1) efficiently mining relevant context from complex configuration files, (2) accurately distinguishing between pre-defined and user-defined parameter values to prevent irrelevant context from being introduced, and (3) managing prompt context overload with iterative, guided interactions with the model. Our evaluations on synthetic and real-world configurations show that CAIP improves misconfiguration detection accuracy by more than 30% compared to partition-based LLM approaches, model checkers, and consistency checkers, uncovering over 20 previously undetected misconfigurations in real-world configurations.

## 1 Introduction

Ensuring the stability, security, and performance of network infrastructures requires carefully configuring routers, switches, firewalls, etc. A misconfiguration on even a single device can lead to black holes, unintended network access, inefficient routes, and a range of other performance and security issues. For instance, a misconfigured packet filter may unintentionally allow unauthorized traffic, exposing the device (and potentially the broader network) to malicious attacks; a buggy routing policy may create loops, thereby degrading performance and rendering local services inaccessible.

Although some configuration changes are automated—especially in large networks [37, 44, 45, 55]—many changes are manually written and verified by domain experts [2]. These experts rely on extensive knowledge gained from years of experience with an organization’s network(s), vast vendor documentation, industry best practices [1], etc. It is especially important for experts to consider the interplay between a configuration change and other aspects of (other) devices’ configurations—*e.g.*, changing a packet filter will impact all interfaces on which the filter is applied. However, considering this interplay is challenging, because even a single device’s configuration is often hundreds or thousands of lines long and contains numerous references to other configuration(s) [7].

To address this challenge, researchers and practitioners have developed numerous automated verification tools. *Model checkers* [3, 5, 6, 17, 21, 41, 46, 51, 62, 65] analyze network-wide forwarding behavior under various conditions (*e.g.*, link failures). These tools require (i) developer-specified structures, algorithms, and/or constraints that codify protocol- and vendor-specific semantics, and (ii) operator-specified policies that codify network-specific forwarding requirements. Both require significant manual effort to define (only some algorithms and policies can be automatically inferred [9, 24, 48]) and both are difficult to define correctly and completely [8, 62]. *Consistency checkers* [16, 22, 23, 26–28, 53], on the other hand, compare configurations across devices or with best-practices and flag deviations as potential issues. While this approach requires minimal manual effort, it lacks semantic depth.

Given these limitations, recent efforts [10, 14, 33, 36, 59, 60] explore the feasibility of using large language models (LLMs) as domain experts to carry out the task of misconfiguration detection. LLMs are pre-trained on extensive corpora, including protocol specifications, vendor documentation, and publicly available configurations. This makes LLMs analogous to human experts with broad domain knowledge. Thus, feeding a configuration under review to an LLM has the potential to identify issues in a fully-automated, semantic-aware manner.

However, feeding the LLM a configuration in its entirety or in fixed-sized partitions, as done by existing tools [10,

[14, 33, 36, 59, 60], has several limitations. Like human engineers, LLMs require *context* to identify issues. Human experts do not examine configurations in a rigid, top-down fashion; instead, they identify relevant configuration lines and dynamically “jump” to related parts of the configuration, regardless of partition boundaries. Similarly, LLMs perform better when provided with concise, relevant context that reflects these interactions. Feeding the entire configuration at once risks exceeding token limits, and splitting configurations into arbitrary partitions can obscure dependencies between different parts. Furthermore, overloading the model with irrelevant or excessive information can dilute its reasoning capabilities, much like a human struggling to recall long-past details.

In this paper, we propose a framework for efficiently extracting and feeding context into LLMs to address these issues and improve misconfiguration detection in routing devices. Specifically, we introduce the *Context-Aware Iterative Prompting* (CAIP) framework, which tackles three key challenges:

**Challenge 1: Automatic and Accurate Context Mining:**

Router configurations are long and complex [7], featuring interdependent lines. Imagine a domain expert tasked with analyzing a configuration file. Instead of reviewing every line exhaustively, they focus on key details and related information. Similarly, automatically extracting and selecting relevant context is essential to reduce computational expense and improve detection accuracy in LLMs. **Solution:** We model the configuration as a hierarchical tree, where each line corresponds to a path from the root to a parameter value. This structure enables systematic identification of three main categories of context—*neighboring*, *similar*, and *referenced* configuration lines—ensuring the LLM receives a concise yet sufficient corpus for analysis.

**Challenge 2: Parameter-Value Ambiguity in Referenceable Context:**

Configurations contain pre-defined and user-defined parameters, with the latter often requiring specialized context for accurate interpretation. Consider a scenario where an expert encounters a user-defined parameter (*e.g.*, an ACL name) and must track its references across the configuration. Similarly, LLMs require clarity in resolving such references. Mislabeling or omitting these references inflates computational costs and increases error risk. **Solution:** We employ an existence-based labeling approach within the configuration tree to distinguish user-defined values by detecting their occurrences as internal nodes in other paths. These classifications are refined with a majority-voting scheme to ensure consistency across contexts.

**Challenge 3: Managing Context Overload in Prompts:**

When faced with an overwhelming volume of irrelevant information, even the best experts can lose focus. Likewise, even after extracting relevant information, the quantity of required context can exceed a single prompt’s capacity. Overloading an LLM with excessive detail reduces coherence and degrades performance. **Solution:** In our iterative prompting framework,

the LLM explicitly requests further context when needed, incrementally refining the prompt and focusing on the most critical lines. This approach mitigates overload and enables more precise detection.

We evaluate CAIP through two distinct case studies. First, we analyze real-world device configuration snapshots, inject realistic updates (including both correct modifications and latent misconfigurations), and demonstrate that CAIP outperforms partition-based LLM approaches, model checkers, and consistency checkers by at least 30% in error detection accuracy. Second, we exhaustively evaluate CAIP on configuration snapshots from a medium-sized campus network. CAIP identifies over 20 previously overlooked misconfigurations, highlighting its practical utility.

## 2 Background and Motivation

Routing device misconfigurations are a common and persistent issue, often leading to security vulnerabilities, performance degradation, or even network outages [16, 64]. Addressing these misconfigurations manually is akin to having a domain expert, such as a network engineer, painstakingly analyze every configuration line. While an expert leverages years of experience and an in-depth understanding of standard practices, the complexity and interdependency of modern configurations [7, 29] make this process not only labor-intensive but also prone to error—especially at scale.

To reduce this reliance on manual effort, automated tools such as model checkers and consistency checkers have emerged. Model checkers [3, 5, 6, 17, 21, 41, 46, 51, 62, 65] identify misconfigurations by creating a logical model of the control and data planes, then analyzing or simulating whether engineer-specified forwarding policies are satisfied. For example, Batfish simulates the control and data planes under specific network conditions (*e.g.*, link failures) and checks whether forwarding paths deviate from specified policies [17]. Similarly, Minesweeper uses a satisfiability solver to verify whether reachability and load-balancing policies hold under various scenarios (*e.g.*, all single link failures) [6]. Although model checkers are popular, their efficacy depends heavily on the correctness and completeness of the predefined policies, which in turn require significant domain expertise to design. Moreover, they often struggle with the nuances of particular protocols and vendor-specific features [8, 62].

In contrast, consistency checkers such as *Diffy* [23] and *Minerals* [27] use statistical methods to learn typical configuration patterns, identifying deviations as potential errors rather than relying on predefined policies. For instance, *Diffy* learns configuration templates to spot anomalies in new configurations, while *Minerals* applies association rule mining to detect misconfigurations by deriving local policies from router files. Unfortunately, these approaches often oversimplify configurations by treating any deviation from standard patterns as misconfigurations—leading to false positives whenever valid, network-specific or device-specific choices differ from

general norms.

Given the limitations of existing tools, an ideal solution for misconfiguration detection would employ an automated, intelligent *agent* with expert-level knowledge of routing device configurations. Such an agent could examine configuration lines in detail and apply domain expertise to detect potential issues. Recent advances in Large Language Models (LLMs) [10, 14, 36, 59, 60] provide a promising avenue toward this goal. Pre-trained on large volumes of network-related data, these models embed a deep understanding of best practices and standards, enabling them to analyze intricate configurations effectively. At the same time, LLMs’ successful applications across diverse domains [12, 13, 19, 38, 40, 49] make their use in network misconfiguration detection a logical next step. A notable example is Ciri [33], an LLM-based configuration verifier that illustrates the potential for scalable and accurate automated detection.

## 2.1 How LLMs Encode and Apply Domain Knowledge

LLMs, particularly transformer-based generative pre-trained models [20, 34, 58], encode domain knowledge during pretraining by processing extensive datasets of network documentation, configuration files, and protocol specifications [4, 11, 43, 57]. Through this process, the models learn statistical patterns, structural hierarchies, and semantic relationships crucial to network operations. For example, they identify how ACL rules typically interact with routing policies or how specific firewall configurations align with best practices. This knowledge is embedded within the model’s parameters, which are refined using gradient descent to minimize prediction errors, effectively transforming the model into a probabilistic repository of domain-specific expertise. Technically, transformers achieve this by employing multi-head self-attention mechanisms, which dynamically assign importance to critical tokens such as protocol keywords or IP ranges, and positional encodings, which preserve the sequence of commands. These mechanisms allow the model to capture both surface-level syntax (*e.g.*, ACL formats) and deeper semantic relationships (*e.g.*, precedence of routing policies over firewall rules). The resulting parameterized knowledge acts like a domain expert’s internalized understanding, enabling the model to generalize its expertise to new configurations it encounters.

This unique architecture allows transformer-based LLMs to fundamentally surpass consistency checkers by integrating pre-trained domain knowledge with context-specific information from the configuration being analyzed, creating a much richer decision-making framework. Consistency checkers rely solely on statistical patterns derived from historical or similar data, using majority-voting logic to flag deviations as anomalies. This approach assumes that frequently observed patterns are always correct and treats all deviations as errors, regardless of whether they represent valid, context-specific configura-

tions. As a result, consistency checkers fail to account for the underlying intent of the configuration or the broader network context, often leading to false positives and missed opportunities to understand nuanced misconfigurations. In contrast, LLMs integrate their pre-learned understanding of best practices with the specific configuration context provided in the prompt. By treating configurations as interconnected systems, they can assess whether deviations align with learned standards or represent genuine misconfigurations. For instance, while a consistency checker might flag a rare ACL rule as an anomaly, an LLM can reason that it is valid based on its relationships to routing policies or protocol directives elsewhere in the configuration.

When a user provides configuration snippets and instructions in a prompt (*e.g.*, “Evaluate this ACL for conflicts”), the LLM combines its pre-trained knowledge with the contextual information in the prompt. The self-attention mechanism enables the model to cross-reference tokens in the prompt with its stored knowledge, such as identifying a conflict between a firewall rule and routing policy based on probabilistic relationships learned during pretraining. By estimating the likelihood of each token or sequence ( $p(\text{token}_i \mid \text{token}_1, \dots, \text{token}_{i-1})$ ), the model highlights deviations from high-probability patterns, flagging potential misconfigurations such as overlapping IP ranges or misplaced `deny` directives. This integration mirrors how a human domain expert draws on years of experience to evaluate new configurations, applying abstract principles to specific cases.

## 2.2 Building Context in Prompts for Accurate Misconfiguration Detection

Through this architecture and the pre-learned general network configuration knowledge, LLM-based Q&A models detect subtle errors that can emerge from interactions among different elements—issues frequently missed by conventional tools. As networks grow more complex and interdependent, the model’s ability to holistically evaluate configurations and understand their intended functions makes it a valuable asset for enhancing misconfiguration detection accuracy. However, to realize this capability, query prompts must include all relevant context from the configuration file when analyzing particular configuration lines or excerpts, including associated segments that might reveal hidden misconfigurations. Incorporating this broader view of the file refines the model’s response by leveraging network-specific context unique to the active configuration, rather than relying solely on pre-learned knowledge. Without this detailed context, the LLM’s capacity to identify potential issues is significantly diminished [25, 35, 50, 56].

**Limitations of Full-File Prompting:** A straightforward solution to providing extensive context is to feed the entire configuration file to the LLM—either as a single prompt or progressively—and let the model detect potential misconfigurations. Yet, this approach is highly inefficient due to token-

length constraints [18, 61, 63]. Moreover, it can trigger *context overload* [30, 32, 42], where extraneous information distracts the model from the crucial lines in the file, leading to missed errors or false positives. Analogously, a human expert would likely be overwhelmed by an avalanche of irrelevant details, making it harder to pinpoint actual issues. Similarly, context overload diminishes the model’s ability to focus and raises the risk of missing important misconfigurations or erroneously flagging innocuous lines.

**Challenges of Partition-Based Prompting:** A common solution to context overload has been partition-based prompting, where the large and complex configuration file is broken down into smaller chunks or sections, typically based on the order in which the configuration appears in the file. Although this method eases token constraints, it also fragments vital context spread across multiple sections. As network configurations often include parameters that interact or depend on other areas of the file, merely analyzing each partition in isolation may overlook key interdependencies. In contrast, a proficient human engineer would cross-reference earlier and later sections continuously; however, a purely partitioned prompting approach discards this global perspective. While prompt chaining [10, 60] can maintain some continuity by linking prompts together, it does not inherently address “context chaining”, and thus crucial interactions between file sections can remain undetected.

Consider a file where firewall rules in one segment conflict with routing policies in another. Partition-based prompting would handle these sections separately, missing the cross-sectional conflict entirely. This fragmentation of context is particularly troublesome for dependency-related misconfigurations, which demand an integrated view of how distinct segments interrelate. If a given chunk of text solely contains lines on unrelated parameters, context about relevant policy sections is lost, rendering the model unable to draw necessary conclusions.

In this paper, we introduce a tool that tackles these issues via a *context-aware, iterative prompting mechanism*. This approach strategically manages and incorporates only the relevant parts of the configuration for each line under review, ensuring that LLMs obtain the required context without succumbing to overload. In the next section, we discuss the specific challenges encountered during the development of this solution and outline our methodology in detail.

### 3 Method

To address the limitations of partition-based prompting, which often fails to capture dependencies across non-adjacent sections of configuration files, we design Context-Aware Iterative Prompting (CAIP), which has two components, as shown in Figure 1: a context mining component and an iterative prompting component. These components work together to automate the process of context extraction and prompt interaction with LLMs, enabling more accurate router mis-

configuration detection.

**1. Context Mining Component:** In this phase, CAIP mines all relevant context for a given configuration line under examination. Relevance here refers to configurations that, while not necessarily directly related, provide important insights, such as neighboring configurations, similar lines applied in different contexts, or referenceable configurations that define key parameters.

**2. Iterative Prompting Component:** Once the relevant context has been mined, the online component engages the LLM through an iterative prompting process. Instead of overwhelming the model with all the extracted context at once, CAIP interacts with the LLM in a guided, sequential manner.

We now explain the challenges encountered in designing these two components and present the solutions.

### 3.1 Context Mining

#### 3.1.1 Challenge 1 - Efficient and accurate context mining

Configuration files are often lengthy and complex, consisting of multiple related and interdependent lines that define various aspects of the behavior of the corresponding network device. When analyzing a specific configuration line, efficiently identifying extracting the relevant context can reduce the computational burden on LLMs and minimize the risk of introducing irrelevant information that could degrade the accuracy of misconfiguration detection. This task is challenging because, although these configurations are typically written in a machine and human-readable format, automating the process of context extraction requires a deep understanding of the hierarchical and interconnected nature of the configuration data. For example, in a network configuration, a line that specifies an access control rule might depend on prior lines that define network segments or user roles. Without properly extracting and including this context in prompt, the LLM might misinterpret the rule, leading to false positives or missed misconfigurations.

To address this challenge, we rely on two observations:

**1. Hierarchical Structure of Configuration Files:** Configuration files are typically written in a structured, hierarchical format that can be effectively modeled as a tree, as shown in the example in Figure 2. In this tree representation ( $T$ ), each node ( $V$ ) corresponds to a specific configuration element, and edges represent the relationships or dependencies between these elements. Let any unique path  $P = \{V_1 \rightarrow V_2(P) \rightarrow V_3(P) \rightarrow \dots \rightarrow V_k(P) = v_k(P)\}$  in this tree have  $k$  nodes, where the value of  $k$  can vary depend on the path taken. The root node ( $V_1$ ) represents the entry point of the configuration file (thus, it functions solely as the root of the tree, providing no semantic meaning, and remains identical regardless of the path taken), while the immediate following node ( $V_2(P)$ ) corresponds to the broadest configuration category in this specific path. Intermediate nodes at subsequent levels ( $V_2(P), V_3(P), \dots, V_{k-1}(P)$ ) represent increas-



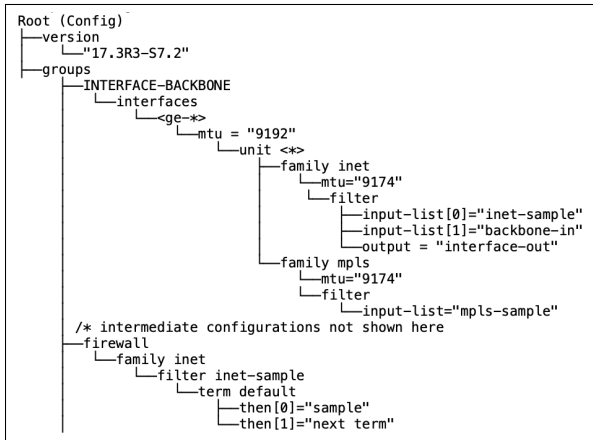
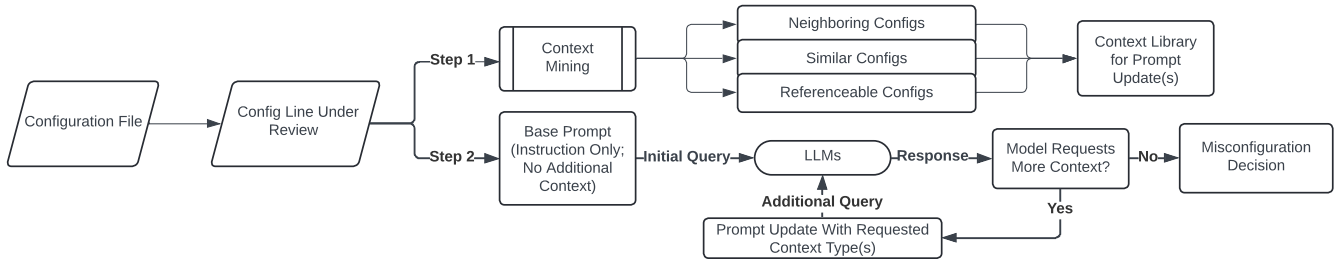


Figure 2: Example snippet of tree-formatted Juniper router configuration file.

ingly nested configuration sections or subcategories, each refining the configuration context. The leaf node  $V_k(P)$  (or parameter node) represents the final configurable parameter in this path, with the associated value  $v_k(P)$ , *i.e.*, parameter values, indicating the specific configuration setting. A unique configuration line is thus a complete path in the tree, starting from the root node and terminating at a parameter node, where the path reflects the hierarchical structure and relationships defined in the configuration file.

**2. Contextual Relevance from Different Aspects:** We observe that the context related to any given configuration line falls into one of three types, based on its position and relation within the configuration file as shown in the example in Figure 3, with increasing levels of complexity to mine. They each providing unique insights that contribute to accurate misconfiguration detection:

*Neighboring Configurations* are closely related to the line under examination, typically within the same section or functional block of the network configuration. For instance, if analyzing a line that defines a VLAN assignment on a particular interface, neighboring configurations would include other lines that configure the same interface or VLAN. These configurations provide insights into how related parameters are set up in proximity, revealing potential dependencies or conflicts. Additionally, by referencing related configurations that define or modify these neighboring parameters, one can

understand how changes in one part of the network might impact adjacent configurations, helping to identify misconfigurations that could lead to issues like traffic misrouting or security vulnerabilities. Neighboring configurations are relatively easy to mine because they are identified based on adjacency or location in the configuration file, making them straightforward to extract.

*Similar Configurations* involve the same type of parameter or function as the configuration line under review but are applied in different contexts within the network. For example, consider configurations that assign IP addresses to different interfaces across various routers in the network. Even though the interfaces and routers may differ, the principles governing IP address assignment remain the same. By comparing these similar configurations, one can detect inconsistencies or deviations from standard practices that might indicate a misconfiguration. This type of context is essential for ensuring that configuration practices are consistent across the network, reducing the risk of errors that could lead to network outages or performance degradation. Mining these configurations is more challenging because, unlike neighboring configurations, they are not located adjacently but must be identified based on functional similarities.

*Referenceable Configurations* provide essential definitions or additional information related to the parameter value being examined. In the network domain, this context is important for understanding how specific values are applied across different parts of the network configuration. For example, if a parameter value specifies an import policy, referenceable configurations might include other lines where this policy is defined or where its behavior is modified. By examining these configurations, one can gain a deeper understanding of how the policy influences routing decisions or interacts with other network elements, ensuring that the configuration is applied correctly and consistently throughout the network. Referenceable configurations are the most difficult to mine because they often involve indirect references and dependencies that are not immediately apparent, requiring a thorough and sometimes recursive analysis to trace how a value or policy is used across various parts of the configuration file.

Given the categorization of relevant contexts and the hierarchical tree model, we can automate the process of mining the aforementioned contexts by translating them into specific

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Example Configuration Line Under Review for Misconfiguration Detection
groups → INTERFACE-BACKBONE → interfaces → <ge-> → unit <*>
→ family inet → filter → input-list = "inet-sample"

Example Extracted Contexts:

Neighboring Configurations
(a) groups → INTERFACE-BACKBONE → interfaces → <ge-> → unit <*>
→ family inet → filter → input-list = "backbone-in"
(b) groups → INTERFACE-BACKBONE → interfaces → <ge-> → unit <*>
→ family inet → filter → output = "interface-out"

Similar Configurations
(a) groups → INTERFACE-BACKBONE → interfaces → <ge-> → unit <*>
→ family mpls → filter → input-list = "mpls-sample"

Referenceable Configurations
(a) groups → firewall → family inet → filter inet-sample → term default
→ then = sample
(b) groups → firewall → family inet → filter inet-sample → term default
→ then = next term

```

Figure 3: Example context mined on selected config line.

paths within the tree. This involves the following steps:

**Neighboring Configuration Mining:** Following the hierarchical tree structure of the configuration file, we can model the entire configuration as a tree  $T$  with nodes representing configuration elements and edges depicting the relationships between them. The root node  $V_1$  represents the entry point of the configuration file and the  $V_2$  nodes represent the broadest configuration categories. For a given configuration path  $P = \{V_1 \rightarrow V_2(P) \rightarrow V_3(P) \rightarrow \dots \rightarrow V_k(P) = v_k(P)\}$  within the tree, where  $v_k(P)$  is the parameter value assigned to the parameter node  $V_k(P)$ , the goal of neighboring configuration mining is to extract other paths in the tree that share a common sub-path with  $P$  up to a certain depth.

The common ancestry level, or shared node depth, can be adjusted to control how much context is mined. Let us define  $m$  as the number of shared nodes from the root, then neighboring configurations are defined as:

$$N_m(P) = \{P' \in T \mid P' = V_1 \rightarrow V_2(P') \dots \rightarrow V_m(P') \rightarrow V_{m+1}(P') \rightarrow \dots \rightarrow V_k(P') = v_k(P')\},$$

where  $P'$  shares the first  $m$  nodes with  $P$ , but the nodes following  $V_m(P')$  (denoted as  $V_{m+1}(P'), V_{m+2}(P'), \dots$ ) can differ from the remaining nodes in  $P$ , and  $P'$  does not end at the same parameter node as  $P$ , i.e.,  $V_k(P') \neq V_k(P)$ .

Adjusting  $m$  allows for control over the size and relevance of the neighboring context. A common choice is to set  $V_m(P) = V_m(P') = V_{k-1}(P)$ , which captures sufficient context while avoiding unnecessary complexity. For example, let the configuration path under review be represented as

$$P = \{V_1 \rightarrow A \rightarrow B \rightarrow C \rightarrow D = v_D\} \mid V_{k-1}(P) = C$$

The set of neighboring paths  $N(P)$  can be defined as:

$$N(P) = \{P' \in T \mid P' = V_1 \rightarrow A \rightarrow B \rightarrow C \rightarrow D' = v_{D'}, D' \neq D\}$$

For example, consider a configuration line that defines the IP address for a particular interface:

$$P = \{V_1 \rightarrow \text{interfaces} \rightarrow \text{ge-0/0/0} \rightarrow \text{unit 0} \rightarrow \text{family inet} \rightarrow \text{address} = 192.168.1.1/24\}$$

Here,  $V_{k-1}(P)$  would be ‘family inet’. Neighboring paths in this case might include:

$$N(P) = \{V_1 \rightarrow \text{interfaces} \rightarrow \text{ge-0/0/0} \rightarrow \text{unit 0} \rightarrow \text{family inet} \rightarrow \text{mtu} = 1500\}$$

This ensures the extracted neighboring paths are closely relevant, revealing how related parameters are configured in the same section without overwhelming the system with excessive data. Thus, in our evaluation of CAIP, we select  $V_{k-1}(P)$  as  $V_m(P)$  for an effective balance between relevance and computational efficiency.

**Similar Configuration Mining:** To identify similar configurations, we focus on paths that share the same root node ( $V_1$ ), broadest category node ( $V_2$ ), and parameter node ( $V_k$ ), but differ in their intermediate nodes or parameter values. Let  $P = \{V_1 \rightarrow V_2(P) \rightarrow V_3(P) \rightarrow \dots \rightarrow V_k(P) = v_k(P)\}$  represent the configuration line under review again. The set of similar configuration paths  $S(P)$  is defined as:

$$S(P) = \{P' : P' = \{V_1 \rightarrow V_2(P') \rightarrow \dots \rightarrow V_k(P') = v_k(P')\}\},$$

where  $P'$  shares  $V_2(P') = V_2(P)$  and  $V_k(P') = V_k(P)$ , but may have different intermediate nodes ( $V_3(P'), V_4(P'), \dots$ ) and a different parameter value  $v_k(P')$ .

By ensuring  $V_2(P') = V_2(P)$ , we compare configurations within the same category or context, even if the paths leading to  $V_k(P')$  and  $V_k(P)$  differ. Using only the root node  $V_1$  would be too broad, as it encompasses the entire configuration file and does not provide meaningful differentiation. This approach allows us to understand how the same type of parameter is configured across different instances, which is crucial for ensuring consistency in network configurations.

For example, let the configuration path under review be:

$$P = \{V_1 \rightarrow \text{Interfaces} \rightarrow \text{Ethernet0} \rightarrow \text{IP} \rightarrow \text{MTU} = 1500\}$$

This represents the configuration for setting the Maximum Transmission Unit (MTU) to 1500 on interface Ethernet0. A similar configuration  $P'$  could be:

$$P' = \{V_1 \rightarrow \text{Interfaces} \rightarrow \text{Ethernet1} \rightarrow \text{IP} \rightarrow \text{MTU} = 9000\}$$

In this case,  $P'$  shares the same broad configuration category ‘Interfaces’ ( $V_2(P') = V_2(P)$ ) and parameter node ‘MTU’ ( $V_k(P') = V_k(P)$ ), but differs in the intermediate node ‘Ethernet1’ and the parameter value (9000 vs. 1500). By comparing similar configurations, the context reveals not only the MTU settings but also the intermediate configuration elements leading to the MTU setting across different interfaces within the network configuration.

**Referenceable Configuration Mining:** In the hierarchical tree structure, referenceable configurations are identified by locating paths where the parameter value of the current line under review appears as an intermediate node in other paths.

This context is crucial for understanding how specific parameter values or policies are further referenced, elaborated upon, or applied in other parts of the configuration. We can formalize this set of referenceable configuration paths as:

$$R(P) = \{P' : P' = V_1 \rightarrow \dots \rightarrow v_k(P) \rightarrow \dots \rightarrow V_k(P')\} = v_k(P').$$

In these paths,  $v_k(P)$  is no longer a parameter value, but an intermediate node, meaning it is being referenced or defined further down the configuration hierarchy.

For example, suppose the current configuration line is:  $V_1 \rightarrow RouterA \rightarrow Policy \rightarrow ImportPolicy = PolicyX$ , a referenceable path might be:  $V_1 \rightarrow RouterA \rightarrow Policy \rightarrow PolicyX \rightarrow Filter = AllowAll$ , indicating that  $PolicyX$  is further applied or modified by the filter configuration. Referenceable configuration mining helps to understand not only how a particular configuration value is defined, but also how it interacts with or influences other components of the network. This process is essential for detecting misconfigurations that arise from improper application or dependency handling across different sections of the configuration file.

One of the most important problems in extracting referenceable configurations is the risk of incorporating irrelevant context, particularly because the process does not distinguish between pre-defined parameter values inherent to the configuration language and custom values defined by network administrators. We now proceed to discuss this problem in more detail and present our solution.

### 3.1.2 Challenge 2 - Referenceable Parameter Value Ambiguity

In network configurations, parameter values have two types: *pre-defined* values and *user-defined* values. Pre-defined values are those that are built into the configuration language itself, such as boolean flags (True vs. False) or access control decisions (Allow vs. Deny). These values are understood by the configuration parser and typically do not require any additional definition or context within the configuration file. On the other hand, user-defined values are customizable by the network administrator and can vary widely depending on the specific requirements of the network. These include values such as IP addresses, timeout intervals, VLAN IDs, and other numeric or alphanumeric identifiers. However, there rarely exists documentation explicitly specifying these types and requires a lot of manual examination, which is hard to scale.

When extracting context for referenceable configurations, failing to differentiate between pre-defined and user-defined values can introduce irrelevant or misleading information into the extracted context. For instance, a pre-defined value like True is universally recognized by the configuration parser and could be used in multiple contexts, such as enabling a feature ( $FeatureX \rightarrow Enabled = True$ ) or setting a protocol flag ( $OSPF \rightarrow PassiveInterface = True$ ). Mining based on this pre-defined value could lead to irrelevant results, pulling in unrelated

configurations that share the same boolean logic, thereby contaminating the context pool. Conversely, user-defined values like IP addresses, a prefix limit (e.g., "maximum-prefixes 500" in a BGP configuration) or a timeout interval (e.g., 30 seconds) are specific to the network and typically set by network operators. These values may appear in routing tables, NAT rules, or timeout policies, with their significance depending on how they're defined and applied. In this case, mining should focus on retrieving all related configurations that define or use these values.

Distinguishing between pre-defined and user-defined values is crucial to avoid extracting irrelevant context, which can lead to several issues:

1. *Context Contamination*: Irrelevant context introduced during the context mining process might erroneously group together configurations that pertain to entirely different aspects of network operation. This contamination of the context pool dilutes the relevance of the mined information, reducing the precision of the LLM and increasing the likelihood of false positives or missed errors.
2. *Reducing Computational Costs*: LLMs can be computationally expensive, especially when processing large-scale network configurations with many interdependent components. Distinguishing between pre and user-defined values can help optimizing the context extraction process, ensuring that only relevant and necessary contexts are passed to the model. This reduces the computational overhead associated with processing extraneous information.

**Existence and Majority-Voting Based Differentiation:** To accurately differentiate between pre-defined and user-defined parameter values within configuration files, we propose a solution that combines existence checks within the configuration tree and a majority-voting mechanism.

1. *Existence-Based Differentiation*: User-defined values often carry contextual information and are typically associated with further definitions or explanations elsewhere in the configuration file. For example, if a parameter value represents a specific, customized import policy, the configuration should contain other lines that elaborate on this policy's behavior. In the hierarchical tree model, if a parameter value is user-defined, it is likely to appear as an intermediate node in other configuration paths, indicating that it is referenced or elaborated upon elsewhere. Conversely, if no such paths exist, the value is likely pre-defined and requires no additional context. Consider the parameter RoutingPolicy with a value of ImportPolicy1 as an example. If ImportPolicy1 appears as an intermediate node in other configuration lines, such as those defining specific route maps or filters, it is likely user-defined. On the other hand, a parameter value like 'PassiveInterface = True' might not have any additional references, indicating that True is a pre-defined value.

**Formal Definitions:** Let  $Val_{user}$  be the set of user-defined values, and  $Val_{pre}$  be the set of pre-defined values. A param-

eter value  $v_k$  is considered *user-defined* if it appears as an intermediate node in at least one other configuration path  $P' \in T$ . This can be expressed as:

$$v_k \in \text{Val}_{\text{user}} \iff \exists P' = \{V_1 \rightarrow \dots \rightarrow v_k \dots \rightarrow V_k(P')\} \in T$$

A parameter value  $v_k$  is considered *pre-defined* if it does not appear as an intermediate node in any other configuration path  $P' \in T$ . This can be formalized as:

$$v_k \in \text{Val}_{\text{pre}} \iff \nexists P' = \{V_1 \rightarrow \dots \rightarrow v_k \dots \rightarrow V_k(P')\} \in T$$

2. *Majority-Voting for Consistency*: A shortcoming with existence-based differentiation is that the same parameter value can be user-defined in one context and pre-defined in another. For instance, the value 1000 used in a timeout setting might be pre-defined and require no further explanation. However, the same value 1000 used as a policy name or group identifier could be user-defined and require contextual elaboration. This distinction is crucial, as identical values can have different meanings depending on the associated configuration parameter. To address this ambiguity, we avoid assigning a universal type to a parameter value across all configurations. Instead, we determine whether a value is pre-defined or user-defined based on the specific combination of the configuration parameter and its associated value. For each configuration parameter, we analyze all associated values. If the majority of these values are user-defined, we classify the entire parameter ( $V_k$ ) as well as all of its possible values ( $v_k$ ) as user-defined. Conversely, if the majority are pre-defined, the parameter is classified accordingly. This approach leverages the principle that configuration parameters should exhibit uniformity in the type of values they accept, maintaining consistency across the configuration.

This further transforms our previous definition of  $\text{Val}_{\text{user}}$  and  $\text{Val}_{\text{pre}}$  into:

$$(V_k, v_k) = \begin{cases} \text{User-Defined, if } \sum_{i=1}^n \mathbb{I}(v_i \in \text{Val}_{\text{user}}) > \frac{n}{2}, \\ \text{Pre-Defined, if } \sum_{i=1}^n \mathbb{I}(v_i \in \text{Val}_{\text{pre}}) \geq \frac{n}{2}. \end{cases}$$

where  $n$  is the number of unique values associated with the configuration parameter  $V_k$ , and  $\mathbb{I}$  is the indicator function that checks whether a value is user-defined or pre-defined.

Example: For the parameter Timeout, where values like 1000, 2000, and 3000 are used, if the majority lack associated definitions, they are treated as pre-defined. Conversely, for a parameter like ImportPolicy, where values such as PolicyA, PolicyB, and 1000 (as a policy name) are used, if most have contextual definitions, all values under that parameter are treated as user-defined.

By combining existence-based checks with a majority-voting mechanism, we can effectively differentiate between

pre-defined and user-defined parameter values in network configurations. This method ensures that the context extracted for LLM analysis is both relevant and accurate, thereby enhancing performance and reducing computational costs. Moreover, this approach maintains consistency within the configuration, ensuring that similar parameters are uniformly treated across different contexts.

## 3.2 Iterative Prompting

As we transition from context mining to utilizing the extracted information in LLM prompts, the next challenge is how to feed this information into the model without overwhelming it. While extracting accurate and relevant context is essential, the model's ability to process that information effectively is just as crucial.

### 3.2.1 Challenge 3 - Context Overload in Prompting

A significant limitation of many LLM-based approaches is context overload, where the model is fed too much information, causing it to lose focus on the most pertinent details. Even with the precise context extraction mechanisms we have, the extracted context can sometimes be extensive, leading to several issues:

1. *Dilution of Relevance*: The model might lose track of the most critical elements of the prompt, resulting in responses that are less accurate or less relevant. For instance, in a complex router configuration, key misconfiguration details could get buried in a flood of less relevant context.
2. *Loss of Coherence and Accuracy*: The model might produce disjointed responses or fail to address core misconfigurations when processing too much information at once. This can lead to fragmented reasoning or errors, especially with intricate router settings. Additionally, token limitations may cause the model to truncate important sections or struggle to process excessive context, further impacting performance.

For example, in detecting routing policy misconfigurations, a single misconfigured line may have dependencies across several sections of the file. Presenting all related context at once can cause the model to fail at prioritizing critical information, leading to missed/incorrect detection. STOA methods, such as prompt chaining, which guides the model through chained instructions, doesn't solve this issue, as it doesn't dynamically adjust or prioritize the context based on model needs.

**Solution:** To address these issues, CAIP implements an iterative, sequential prompting framework that allows the model to request and process relevant context in stages. Instead of overwhelming the model with all available context at once, CAIP enables the model to actively request additional information as needed. This approach enhances the model's ability to maintain focus, coherence, and overall detection accuracy by allowing it to prioritize and process context more effectively, ensuring that only the most pertinent information is presented at each step.



```

Initial Prompting

"I have a line of network route configuration for a Juniper router and
need to determine if there is any misconfiguration:
Configuration Line Under Review: groups-->INTERFACE-BACKBONE-->interfac
es--><xe-->-->mtu=True
Given this information, please now proceed to analyze the provided con
figuration to identify any potential SYNTAX misconfigurations.
Do not hallucinate if no misconfiguration is occurring.
Additional contextual information with respect to the configuration li
ne under review can be provided upon request if useful for identifying
SYNTAX misconfigurations, include (1) Neighboring Configu
ration Lines; (2) Configuration Lines with Similar Pattern and Nature;
(3) Referenceable Configuration Lines; (4) Referenceable Configuration
Lines for Neighboring Configuration Line.
Respond in a json format strictly adhering to the following and do not
provide any other output:
If no further contextual information is needed:
{
  "Misconfigured": boolean, // true if there is any misconfiguration
  "Parameter": [], // List of misconfigured parameters
  "reason": [] // List of misconfiguration explanations
}
If further contextual information is needed:
{
  "requestedInfo": [], // List containing additionally requested
    contextual information, select from [1,2,3,4]
  "reason": [] // List of explanations for requestedInfo
}"

Example LLM Response for Context Request

{
  "requestedInfo": [1, 2],
  "reason": ["Neighboring configuration lines are necessary to confirm
    the syntax structure in context.", "Configuration lines
    with a similar pattern and nature are needed to ensure
    consistent parameter usage."]
}

```

Figure 4: Example: initial prompting and LLM context request response for detecting syntax misconfiguration.

**1. Initial Prompting:** The process begins by feeding the LLM the specific configuration line under review, along with targeted instructions about the type of misconfiguration we are checking for (*e.g.*, syntax errors, dependency conflicts, or just general misconfiguration). Figure 4 presents an example where the initial prompt includes MTU configuration line with an instruction to identify any syntax misconfigurations related to that configuration. This reduces the initial information load and allows the model to focus on the core issue.

**2. Contextual Options:** Next, CAIP offers the model the option to request additional context based on its understanding of the configuration line and the misconfiguration detection request. These options are based on our categorized context types, including: neighboring ( $N(P)$ ), similar ( $S(P)$ ), and referenceable contexts ( $R(P)$ ), as well as referenceable contexts on neighboring configurations ( $N(R(P))$ ). By allowing the model to choose which context to receive, CAIP ensures that the LLM is only given information that is most relevant to the detection task at hand, reducing the risk of context overload. Figure 4 illustrates an example request response where the LLM requests neighboring context and referenceable contexts on neighboring configurations after the initial prompt to aid analyzing the MTU setting.

**3. Iterative Refinement:** After the model processes the initial prompt, it can request additional information if needed before arriving at the final decision. CAIP engages in a feedback loop where the model’s output informs the next set of prompts. For instance, if the model identifies a possible syntax issue but requires more context from a related policy definition, *i.e.*, similar context, CAIP can deliver that specific context in the next prompt. This iterative process continues until the model

```

Additional Context Added to Prompt

Neighboring Configuration Lines:
- groups-->INTERFACE-BACKBONE-->interfaces--><xe-->-->unit <*>
  - family inet-->filter-->input-list-->1=backbone-in
  - family inet-->filter-->input-list-->0=uplink-in
  - family inet-->filter-->output=interface-out
  - family iso-->mtu=1497
  - family mpls-->filter-->input=mpls-sample
  - family inet6-->filter-->output=v6filter
  - family inet6-->mtu=9174
  - family inet6-->filter-->input-list-->0=inet6-sample
  - family inet6-->filter-->input-list-->1=v6filter
  - family mpls-->mtu=9174
  - family inet-->mtu=9174

Similar Configuration Lines:
- groups
  - INTERFACE-BACKBONE-->interfaces--><ae*>-->unit <*>-->family iso-->
    mtu=1497
    (Further definition DO NOT EXIST for the value [1497] assigned
    to the parameter [mtu] in the rest of the configuration.)
  - INTERFACE-CONNECTOR-->interfaces--><ae*>-->mtu=9192
    (Further definition DO NOT EXIST for the value [9192] assigned
    to the parameter [mtu] in the rest of the configuration.)

Final LLM Response

{
  "Misconfigured": true,
  "Parameter": ["mtu"],
  "reason": ["The value 'True' assigned to the parameter 'mtu' is not
    valid. The 'mtu' parameter typically requires a numeric
    value representing the maximum transmission unit size."]
}

```

Figure 5: Example: Adding requested context and retrieving misconfiguration detection result.

has sufficient information to make a well-informed decision. Figure 5 provides an example of what a final misconfiguration decision looks like regarding a misconfigured MTU value.

This sequential, interactive approach mirrors how a network administrator might manually review a configuration file—examining the most relevant sections first, then digging deeper into referenceable or adjacent configurations as needed.

### 3.2.2 Why Iterative, Requested Based Prompting Works

By breaking down the context into smaller, more digestible portions, CAIP addresses the fundamental challenges of context overload. This method significantly enhances the model’s ability to:

1. Stay on target: With less irrelevant information, the model can maintain focus on the specific issue under review.
2. Preserve coherence: Incrementally adding information allows the model to build a more coherent understanding of the configuration [31, 52], improving detection of complex issues such as dependency conflicts or misapplied policies.
3. Optimize performance: Avoiding context overload reduces the processing burden on the model, leading to faster and more reliable outputs.

For example, in detecting misconfigurations in a multi-layer firewall policy, CAIP might first present the core rule in question, then iteratively offer neighboring rules that affect the same traffic flow, followed by referenceable configuration sections that define broader network policies. This structured process ensures that the LLM has all the information it needs, without drowning in unnecessary details, enabling it to make accurate decisions. Unlike conventional prompt-chaining, which provides incremental instructions based on a fixed context, CAIP dynamically adds context as requested by the LLM, focusing on the evolving needs of the analysis.

## 4 Evaluation

To evaluate the effectiveness of CAIP, we present two test scenarios as case studies and compare CAIP with state-of-the-art solutions. These include:

1. *Synthetic Misconfiguration Verification*: Synthetic misconfigurations are introduced into real network configuration snapshots. We run CAIP on specific configurations before and after the misconfigurations are introduced to evaluate the model’s ability to accurately infer and detect misconfigurations.
2. *Comprehensive Real Configuration Snapshot Verification*: CAIP is run on configuration snapshots from a medium-scale campus network to detect overlooked misconfigurations in a real-world setting. This covers both *targeted* misconfiguration detection, where we focused on known issues like VLAN assignment errors, and *non-targeted* detection, which involved scanning for general misconfigurations without prior knowledge of the issues.

We now introduce the setup for our evaluation and present the details of the case studies in the subsequent sections.

### 4.1 Evaluation Setup

In this section, we describe our evaluation setup.

#### 4.1.1 Context Extraction Hardware Setup

*System Information*: GNU/Linux Red Hat Enterprise Linux release 8.8 (Ootpa); *CPU Specifications*: Architecture: x86\_64; CPU(s): 32 (2 threads/core, 16 cores/socket, 1 socket); Model: AMD EPYC 7302P 16-Core Processor; Max MHz: 3000.000; L3 cache: 16384K; NUMA node0 CPU(s): 0-31

We opt not to use the GPU, as the computational cost of the context extraction and processing tasks was manageable on the CPU — the tasks primarily benefit from parallel CPU threads, and GPU acceleration would not provide significant additional speedup.

#### 4.1.2 LLM Model Used: GPT-4o

For running the LLM-based detection, we use GPT-4o [39], OpenAI’s most recent transformer-based model designed for complex multi-step tasks.

**Specifications**: Model: GPT-4o-2024-05-13; Context window: 128,000 tokens; Max output tokens: 4,096 tokens; Training data Up to October 2023.

#### 4.1.3 Prompting and Query Setup

We use OpenAI’s Chat Completions API to interact with GPT-4o. The system was provided with a structured conversation history to maintain context across multiple queries. For each query, the model was tasked with analyzing the configuration lines to detect potential misconfigurations, with instructions tailored to focus on specific misconfiguration types (e.g., syntax issues, policy conflicts) or general misconfigurations. Figures 4 and 5 show example prompting, queries, and responses.

| Type   | Description            | Misconfig                                    | Requested Context           |
|--------|------------------------|----------------------------------------------|-----------------------------|
| Syntax | Missing brace          | ...interfaces: { "<ge-0/0/0";                | None                        |
|        | Invalid keyword        | ...mtu: "True"                               | $N(P)$ , $R(P)$             |
|        | Incorrect hierarchy    | ...host-name: "[rtsw.alba-re1]"              | $N(P)$                      |
|        | Invalid IP address     | ...neighbor: 192.168.253.1.1"                | $N(P)$ , $R(P)$             |
| Range  | Invalid MTU            | ...mtu: "10000"                              | $N(P)$ , $R(P)$             |
|        | Invalid VLAN ID        | ...vlan-id: "5000"                           | $N(P)$ , $R(P)$ , $N(R(P))$ |
|        | Invalid AS             | ...autonomous-system: 70000                  | $N(P)$ , $R(P)$             |
|        | Invalid prefix limit   | ...maximum: "200000"                         | $N(P)$ , $R(P)$             |
| D/C    | Non-existent group     | ...apply-groups: "L2-LSP-ATTRIBUTES"         | $N(P)$ , $R(P)$ , $N(R(P))$ |
|        | Policy Conflict        | ...community add 12CLOUD-EXTENDED-TARGET     | $N(P)$ , $R(P)$ , $N(R(P))$ |
|        | Non-existent filter    | ...input-list: "uplink"                      | $N(P)$ , $R(P)$             |
|        | Non-existent policy    | ...export: "OECS-400-300-LOOP"               | $R(P)$                      |
|        | Incorrect Filter Usage | ...family mpls: filter:input-list:"v6filter" | $R(P)$ , $S(P)$             |
|        | Disabled Sampling      | ...family inet6: SAMPLING MISSING/DISABLED   | $R(P)$ , $N(P)$             |
|        | VRF Target Conflict    | ...vrf-target target:11537:"313001"          | $S(P)$                      |
|        | Abnormal Small MTU     | ...family iso: mtu: "1497"                   | $R(P)$ , $S(P)$             |
|        |                        |                                              |                             |

Table 1: Synthetic Misconfigurations Introduced and LLM Requested Context for CAIP.

### 4.2 Case Study 1: Synthetic Misconfiguration Verification

We first evaluate CAIP on synthetically introduced misconfigurations, which allows us to systematically test various misconfiguration types and assess detection accuracy in a controlled environment. We broadly categorize router misconfigurations into three types:

- *Syntax errors* occur when the configuration does not adhere to the expected format or structure—e.g., a missing bracket or misused keyword in a BGP routing policy.
- *Range violations* involve parameter values that fall outside the acceptable range—e.g., an MTU value that exceeds the maximum allowed for a specific interface type.
- *Dependency/conflict (D/C) issues* arise when different configuration lines are incompatible or contradict each other—e.g., a firewall rule might block traffic that another policy explicitly permits.

To evaluate CAIP, we obtain snapshot configurations from Internet2 (Juniper devices) and sampled 16 correct configuration lines. These were then modified to introduce synthetic errors across the three misconfiguration types. For syntax and range errors, we created four distinct misconfigurations each, totaling 8 errors (see Table 1). For D/C misconfigurations, we introduced 8 distinct errors, as vanilla LLMs excel at detecting syntax and range issues, while CAIP’s context mining is particularly effective at uncovering nuanced D/C issues. The additional D/C errors allow us to thoroughly assess this capability. For each misconfiguration, we run CAIP by following the two components: (1) treating the misconfigured line as the line under review and extracting all relevant context, and (2) conducting the iterative, sequential prompting process against the model, obtaining the final misconfiguration detection decision.

When forming the prompt, we explicitly instruct the model to look for each type of misconfiguration—syntax, range, or dependency/conflict—individually. This is because the model may request different types of context depending on the specific misconfiguration type it is trying to detect. We report only the results corresponding to the actual misconfiguration type introduced. Importantly, CAIP never yielded false positives when the model was instructed to find a misconfiguration type different from the actual type. Additionally, we find that

asking the model to search for ‘GENERAL’ misconfigurations also led to successful detection, though more context was often requested by the model.

We compare CAIP to three representative tools: the model checker *Batfish* [17], the consistency checker *Diffy* [23], and the partition-based GPT Q&A model *Ciri* [33] using GPT-4o for fair comparison.

| Type                      | Cases     | Number of Corrected Detected Cases |                      |                      |                    |
|---------------------------|-----------|------------------------------------|----------------------|----------------------|--------------------|
|                           |           | CAIP                               | Ciri                 | Batfish              | Diffy              |
| Syntax Error              | 4         | 4/4 (100%)                         | 2/4 (50%)            | 3/4 (75%)            | 0/4 (0%)           |
| Range Error               | 4         | 4/4 (100%)                         | 2/4 (50%)            | 0/4 (0%)             | 0/4 (0%)           |
| D/C Error                 | 8         | 8/8 (100%)                         | 1/8 (12.5%)          | 2/8 (25%)            | 0/8 (0%)           |
| Original Confs (No Error) | 16        | 16/16 (100%)                       | 16/16 (100%)         | 16/16 (100%)         | 16/16 (100%)       |
| <b>Total</b>              | <b>32</b> | <b>32/32 (100%)</b>                | <b>21/32 (65.6%)</b> | <b>21/32 (65.6%)</b> | <b>16/32 (50%)</b> |

Table 2: Synthetic Misconfiguration Detection Results.

The results in Table 2 demonstrate that CAIP outperforms the other tools in detecting all three types of misconfigurations. It achieves a perfect 16/16 detection rate across the board for misconfigurations. This indicates that CAIP’s comprehensive context mining and iterative prompting process effectively identify errors in network configurations, even in more complex cases like D/C issues. In contrast, the other tools showed limitations, particularly when it came to detecting range violations and D/C issues.

**Comparison Against Partition-based LLMs** The partition-based GPT model *Ciri* performs reasonably well in detecting syntax and range violations, with a detection rate of 50% for both categories. This success is due to the LLM model’s training on vast, diverse text, including network-related documentation and configuration files, enabling it to recognize common syntax structures and numerical limits typically found in configurations. Transformer-based LLMs, with their self-attention mechanism, are particularly effective at identifying these common, localized issues. However, it lags significantly behind CAIP in detecting D/C issues, with a 12.5% detection rate in this category. The primary reason for this underperformance is the partitioning approach. This approach analyzes configuration sections in isolation, failing to capture the complex interdependencies between configuration lines, which is crucial for detecting D/C issues. The only D/C misconfiguration that *Ciri* successfully detects is the ‘VRF Target Conflict,’ where the conflicting ‘route-distinguisher’ value configuration happens to be in the same prompt partition, providing just enough context for accurate detection.

As shown in Table 1 and 3, D/C issues, such as policy conflicts, require the model to understand and analyze interactions between different configuration elements. CAIP addresses this by guiding the LLM to request the relevant contextual information it needs for accurate reasoning and detection. The example CAIP-enable LLM responses in Table 3 clearly show that these contexts are being leveraged to derive the correct detection results. In contrast, partition-based GPT struggles to detect such misconfigurations effectively, as it analyzes configurations in isolation and can often miss these

complex interdependencies.

| Type   | Misconfig              | errParameter        | Reason                                                                                                                                                                                                                                                                                   |
|--------|------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Syntax | Missing brace          | ["<ge-0>"]          | "MTU value for 'family inet' is set to 9174 which is less than the interface MTU value of 9192. The MTU for 'family inet' should not be less than the interface MTU."                                                                                                                    |
| Range  | Invalid VLAN ID        | ["maximum"]         | "The 'maximum' value of 200000 in the configuration line under review significantly exceeds the 'maximum' values of 1000 found in similar configuration lines, indicating a potential range misconfiguration."                                                                           |
| D/C    | Policy Conflict        | ["remove-comms-rt"] | "There is a conflict in the configuration under term 'remove-comms-rt' where the action 'community add I2CLOUD-EXTENDED-TARGET' is followed by 'community delete I2CLOUD-EXTENDED-TARGET' which are contradictory actions."                                                              |
|        | Disabled Sampling      | ["inet6"]           | "The family mpls is incorrectly using the IPv6 filter inet6-sample v6filter, which is intended for family inet6 in similar configuration lines. MPLS traffic should have its own specific filter, as the filtering requirements for MPLS are different from those for IPv6."             |
|        | VRF Target Conflict    | ["target"]          | "The route-distinguisher (RD) in neighboring configurations is set to 11537:381841, while the vrf-target is set to target:11537:313001. These values should typically align, or at least follow certain conventions for mapping VRFs to the correct target instances in MPLS/BGP L2VPN." |
|        | Incorrect Filter Usage | ["input-list"]      | "The family mpls is incorrectly using the IPv6 filter inet6-sample v6filter, which is intended for family inet6 in similar configuration lines. MPLS traffic should have its own specific filter, as the filtering requirements for MPLS are different from those for IPv6."             |

Table 3: Example LLM Reasoning for CAIP.

**Comparison Against Non-LLM-based Tools** *Batfish* performs well on syntax errors (75% detection rate) but struggles with range violations and only partially detected D/C issues (2/8). *Batfish* uses a rule-based model checker, which excels in detecting predefined syntax errors but is less effective in handling misconfigurations that involve contextual nuances, such as range violations or dependency conflicts. Specifically, *Batfish* detects issues related to non-existent groups and policies (hard-coded into its rule checks) but missed more subtle D/C misconfigurations, like policy conflicts or filter dependencies.

*Diffy* performs poorly across all misconfigurations, with a 0/16 detection rate, due to its data-driven approach, which focuses on learning common usage patterns from configurations and flagging anomalies as potential bugs. While effective for detecting clear deviations from standard patterns in other configuration files, complex issues like policy conflicts or range violations may not deviate from typical patterns but still be incorrect, which *Diffy*’s template-based anomaly detection cannot capture. Furthermore, *Diffy*’s reliance on identifying deviations from learned patterns makes it less effective at detecting syntax errors and range violations. These types of misconfigurations often follow standard formatting and numerical values that may not stand out as anomalies, making them difficult for *Diffy* to detect without a granular understanding of configuration structure and constraints.

Finally, despite varying performance in detecting misconfigurations, all tools, including CAIP, *Batfish*, *Diffy*, and the partition-based GPT model *Ciri*, successfully marked all correctly configured lines as valid prior to the introduction of the synthetic errors, indicating strong capabilities in avoiding false positives for correctly functioning configurations.



### 4.3 Case Study 2: Comprehensive Real Configuration Snapshot Verification

To evaluate CAIP on real-world misconfiguration detection, we obtain Aruba router configuration snapshots from a medium-scale campus network. We have two objectives: (1) to verify the scalability of CAIP when applied to larger network configurations, and (2) to investigate whether CAIP can detect potential misconfigurations that have not yet been identified by existing tools.

We perform two types of analysis, beginning with *targeted misconfiguration detection*: A key aspect of this case study is the demonstration of CAIP’s flexibility in integrating additional context types under different scenarios. This is particularly useful when network operators have prior domain knowledge about specific misconfigurations they are trying to detect that fall outside of CAIP’s default context library. To illustrate this flexibility, we consult the network operators who provided the configuration snapshots. They were specifically interested in detecting incorrect ‘VLAN assignments’ — a type of misconfiguration that often requires a network-wide view. Such misconfigurations are usually identifiable only when considering the configurations of other devices in the same network, as deviations from uniform configurations can indicate potential issues. To address this, we introduce an additional, default context type, called ‘Intra-Router Consistency Context’. This context type mines the prevalence of the same parameter-value pair across other devices in the network, providing insights into whether a configuration is common or potentially erroneous. For this scenario, we modify the initial prompt to focus specifically on detecting ‘WRONG VLAN ASSIGNMENT’.

Example Intra-Router Consistency Context extracted:

*‘For the Configuration Line Under Review, the same configuration is found in 189 out of 191 other configuration files. (Significantly lower prevalence may indicate an uncommon or potentially erroneous configuration.)’*

The second analysis involves *non-targeted misconfiguration detection*, where CAIP uses its default context library to aim the LLM for detecting general misconfigurations without prior knowledge of specific issues. In this mode, no specific misconfiguration type is provided to the model; instead, CAIP exhaustively analyzes the entire configuration file to identify ‘GENERAL’ errors. This approach is particularly valuable for detecting overlooked misconfigurations in large, complex network environments where errors may not be immediately obvious or anticipated. By running CAIP in this mode, we assess its ability to detect subtle misconfigurations across the network without predefined expectations, ensuring it can operate effectively even when network operators are unsure of the specific issues they might face.

We perform exhaustive analysis using CAIP across 11 configuration files covering  $\sim 6\%$  of all devices on the campus network, applying the context mining framework to extract all

| Analysis Type | Misconfigs                  | Count | Severity (Reason)                                                                                | True Positive Rate (TPR) |
|---------------|-----------------------------|-------|--------------------------------------------------------------------------------------------------|--------------------------|
| Targeted      | Wrong VLAN Assignment       | 6     | High: Wrong VLANs permitted in configuration, causing access issues.                             | 2/6 (33.3%)              |
| Non-Targeted  | Invalid Subnet Mask         | 2     | High: Subnet mask usage, e.g., 255.255.253.0 violating binary boundary rules.                    | 2/2 (100%)               |
|               | Inconsistent Policy Naming  | 14    | Low: Misnamed ‘snmpread’ for ‘snmp_communities’ but with consistent usage.                       | 14/14 (100%)             |
|               | Insecure SSH KEX Algorithm  | 1     | Low: Insecure ‘diffie-hellman-group14-sha1’ used (low risk as stronger algorithms are included). | 1/1 (100%)               |
|               | Ambiguous Rate Limit Values | 2     | Ambiguous: ‘Interval’ and ‘burst’ set to 0, leading to nondeterministic behaviors.               | 2/2 (100%)               |

Table 4: Comprehensive Real Configuration Snapshot Verification Results: CAIP-Detected Misconfigurations

relevant context for each configuration line. We then prompt the model to identify the corresponding misconfigurations, allowing it to dynamically request the necessary context during the iterative prompting phase. The detection results are presented in Table 4, highlighting CAIP’s ability to uncover new misconfigurations. For each of misconfigurations detected by CAIP, we verify the correctness of the inference decision with domain experts.

**Targeted Analysis** In the targeted analysis, CAIP demonstrates consistent performance in identifying misconfigurations related to incorrect VLAN assignments. Out of the six misconfigurations detected, CAIP correctly flags two with a true positive rate (TPR) of 33.3%. The identified misconfigurations involved wrong VLAN assignments that caused misrouting and access issues, a high-severity problem for network operations.

Despite the modest TPR, the false positives align with the domain experts’ expectations. Network experts clarified the VLAN assignment deviations falsely flagged as misconfigurations are in fact intentional modifications tailored to the specific routers within the network. In this regard, we compare CAIP’s detection results (both true and false positives) with those from the experts’ internal tool—a graph-based, non-LLM solution designed to check intra-router consistency—and the results are consistent. Thus, while the false positives may appear significant, they represent deviations from standard configurations that would typically signal errors, but in these cases, are expected and intentional adjustments.

**Non-Targeted Analysis** In the non-targeted analysis, CAIP was able to detect several types of misconfigurations across the provided configuration snapshots, as shown in Table 4. The detected misconfigurations were categorized by severity based on their potential impact on network operations:

- **High Severity:** Misconfigurations in this category pose critical risks to the network and require immediate action — CAIP identified invalid subnet mask (e.g., 255.255.253.0) usages that violate binary boundary rules, which could lead to routing issues and network instability.
- **Low Severity:** These issues may not immediately disrupt network operations but should be addressed to ensure optimal functionality and avoid potential risks— CAIP identified many instances of misnamed configurations (e.g., ‘snmpread’ instead of ‘snmpread’ for ‘snmp\_communities’) and inclusion of insecure SSH KEX algorithm which (‘diffie-hellman-group14-sha1’), do not break the system but could

lead to confusion and maintenance challenges.

- *Ambiguous*: These misconfigurations are related to configurations that lead to undefined or nondeterministic behaviors — CAIP found ambiguous rate limit values (‘interval’ and ‘burst’ set to 0), which can cause unpredictable performance issues due to missing documentation on how these values should be handled, suggesting updates or further specifications in the documentation are needed.

Specifically, CAIP successfully identified 19 misconfigurations, including two cases of invalid subnet masks, 14 instances of inconsistent policy naming, one insecure SSH Key Exchange (KEX) algorithm, and two ambiguous rate limit values. Validated with domain experts, we find all misconfiguration cases are either valid or justifiable with a TPR of 100% across all targeted misconfigurations.

Across synthetic and real-world configurations, CAIP consistently outperforms state-of-the-art tools like Batfish, Diffy, and partition-based GPT (Ciri), especially in detecting complex misconfigurations. CAIP’s dynamic context mining and iterative prompting allow it to capture interdependencies that other methods missed. Both in targeted and non-targeted scenarios, it demonstrates capability in identifying misconfigurations that traditional tools overlooked, making it a robust solution for network configuration validation.

## 5 Discussion and Future Work

CAIP has proven effective in improving the detection of router misconfigurations, but there are several avenues for further exploration and refinement. One challenge is expanding the framework to better handle large-scale, distributed networks with interdependent routers. These environments often involve complex relationships between devices, and future work could focus on extending CAIP to capture and analyze multi-device contexts by default. Incorporating cross-router consistency checks and advanced network-wide policy validation could be valuable enhancements.

Another area for improvement is enhancing the LLM’s ability to process real-time configuration changes. As networks evolve dynamically, incorporating real-time monitoring data into the context mining process would allow for continuous verification and faster misconfiguration detection. Integrating CAIP with network management tools that track operational metrics could provide valuable runtime data, further improving detection accuracy by correlating live data with configuration snapshots.

Future enhancements can also involve proposing and implementing automated fixes for detected misconfigurations with real-time updates. This would help network operators quickly address issues, streamlining troubleshooting and repairs.

## 6 Conclusion

In this paper, we introduced CAIP, a framework for router misconfiguration detection that leverages context-aware itera-

tive prompting to improve LLM accuracy. By systematically extracting relevant context from network configurations and following a guided, interactive prompting mechanism, CAIP addresses limitations in current model checkers, consistency checkers, and LLM-based models. Through case studies involving both synthetic and real-world misconfigurations, we demonstrated that CAIP outperforms existing methods across different misconfiguration types, ensuring more reliable and scalable network configuration verification.

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